

1

```

10 • SELECT e.emp_id,e.emp_name,d.dept_name,d.location FROM employees e
11     INNER JOIN departments d ON e.dept_id = d.dept_id;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	emp_id	emp_name	dept_name	location
▶	101	Arjun	Engineering	Chennai
	102	Priya	Engineering	Chennai
	107	Sanjay	Engineering	Chennai
	103	Karthik	Sales	Hyderabad
	106	Divya	Sales	Hyderabad
	109	Vikram	Sales	Hyderabad
	104	Meena	HR	Bengaluru
	108	Anitha	Research	Pune

2

```

9
10 • SELECT e.emp_id,e.emp_name,d.dept_name FROM employees e
11     LEFT JOIN departments d ON e.dept_id = d.dept_id;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	emp_id	emp_name	dept_name
▶	101	Arjun	Engineering
	102	Priya	Engineering
	103	Karthik	Sales
	104	Meena	HR
	105	Rahul	NULL
	106	Divya	Sales
	107	Sanjay	Engineering
	108	Anitha	Research
	109	Vikram	Sales

3

```

10 • SELECT d.dept_id, d.dept_name, d.location FROM departments d
11 LEFT JOIN employees e ON d.dept_id = e.dept_id WHERE e.emp_id IS NULL;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	dept_id	dept_name	location
▶	50	Finance	Mumbai

4

```

9
10 • SELECT e.emp_name,p.project_name,d.dept_name,ep.hours FROM employees e
11 INNER JOIN employee_project ep ON e.emp_id = ep.emp_id
12 INNER JOIN projects p ON ep.project_id = p.project_id
13 INNER JOIN departments d ON p.dept_id = d.dept_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	emp_name	project_name	dept_name	hours
▶	Arjun	Banking App	Engineering	120
	Priya	Banking App	Engineering	100
	Sanjay	Banking App	Engineering	90
	Priya	Cloud Migration	Engineering	80
	Sanjay	Cloud Migration	Engineering	130
	Karthik	Sales Portal	Sales	110
	Divya	Sales Portal	Sales	70
	Meena	Employee System	HR	60
	Anitha	AI Research	Research	150

5

```

10 • SELECT p.project_id, p.project_name, e.emp_id,e.emp_name FROM projects p
11 LEFT JOIN employee_project ep ON p.project_id = ep.project_id
12 LEFT JOIN employees e ON ep.emp_id = e.emp_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	project_id	project_name	emp_id	emp_name
▶	201	Banking App	101	Arjun
	201	Banking App	102	Priya
	201	Banking App	107	Sanjay
	202	Sales Portal	103	Karthik
	202	Sales Portal	106	Divya
	203	AI Research	108	Anitha
	204	Employee System	104	Meena
	205	Cloud Migration	102	Priya
	205	Cloud Migration	107	Sanjay
	206	Audit Automation	NULL	NULL

6

```

10 • SELECT e.emp_name AS employee_name, m.emp_name AS manager_name FROM employees e
11 LEFT JOIN employees m ON e.manager_id = m.emp_id;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
employee_name	manager_name		
Arjun	NULL		
Priya	Arjun		
Karthik	Divya		
Meena	NULL		
Rahul	Arjun		
Divya	NULL		
Sanjay	Arjun		
Anitha	NULL		
Vikram	Divya		

7

```

9
10 • SELECT e.emp_id, e.emp_name, COUNT(ep.project_id) AS total_projects FROM employees e
11 INNER JOIN employee_project ep ON e.emp_id = ep.emp_id
12 GROUP BY e.emp_id, e.emp_name HAVING COUNT(ep.project_id) > 1;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
emp_id	emp_name	total_projects	
102	Priya	2	
107	Sanjay	2	

8

```

9
10 • SELECT d.dept_name, COUNT(e.emp_id) AS employee_count, COALESCE(AVG(e.salary), 0) AS average_salary FROM departments d
11 LEFT JOIN employees e ON d.dept_id = e.dept_id
12 GROUP BY d.dept_id, d.dept_name;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
dept_name	employee_count	average_salary	
Engineering	3	75666.666667	
Sales	3	62666.666667	
HR	1	60000.000000	
Research	1	78000.000000	
Finance	0	0.000000	

9

```

10 • SELECT e.emp_name AS employee_name,ed.dept_name AS employee_department,p.project_name,pd.dept_name AS project_department
11 FROM employees e
12 INNER JOIN employee_project ep ON e.emp_id = ep.emp_id
13 INNER JOIN projects p ON ep.project_id = p.project_id
14 INNER JOIN departments ed ON e.dept_id = ed.dept_id
15 INNER JOIN departments pd ON p.dept_id = pd.dept_id
16 WHERE e.dept_id <> p.dept_id;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
employee_name	employee_department	project_name	project_department

10

```

9
10 • SELECT p.project_id,p.project_name,d.dept_name,p.budget FROM projects p
11 INNER JOIN departments d ON p.dept_id = d.dept_id
12 LEFT JOIN employee_project ep ON p.project_id = ep.project_id
13 WHERE ep.emp_id IS NULL;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
project_id	project_name	dept_name	budget
▶ 206	Audit Automation	Finance	350000.00

Code:

1 .Frog Jump 1

```
import java.util.*;
```

```
public class froGJump1 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int n=sc.nextInt();
```

```
        int[] arr=new int[n];
```

```
        for(int i=0;i<n;i++) arr[i]=sc.nextInt();
```

```
        int[] dp=new int[n];
```

```
        if(n==2){
```

```

        System.out.println(Math.abs(arr[1]-arr[0]));
        return;
    }
    dp[0]=0;
    dp[1]=Math.abs(arr[1]-arr[0]);
    for(int i=2;i<n;i++){
        dp[i]=Math.min(dp[i-1]+Math.abs(arr[i]-arr[i-1]),dp[i-2]+Math.abs(arr[i]-arr[i-2]));
    }
    System.out.println(dp[n-1]);
}
}

```

2. Frog Jump 2

```

public class froGJump2 {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] arr=new int[n];
        int k=sc.nextInt();
        for(int i=0;i<n;i++) arr[i]=sc.nextInt();
        int[] dp=new int[n];
        Arrays.fill(dp,Integer.MAX_VALUE);
        dp[0]=0;
        for(int i=1;i<n;i++){
            for(int j=1;j<=k && i-j>=0;j++){
                dp[i]=Math.min(dp[i],dp[i-j]+Math.abs(arr[i]-arr[i-j]));
            }
        }
    }
}

```

```
    }  
  }  
  System.out.println(dp[n-1]);  
}  
}
```